9.4 Toward a Category of Polytopes ?

Fiber polytopes form an important first step in investigating a "category of polytopes," a program suggested by Louis Billera. In fact, this category should have interesting properties that are fundamental to many geometric questions. It is surprising that the basic "universal constructions" for such a category have hardly been studied. Among them are the *fiber polytopes*, which form "kernel objects"; the *mapping polytopes* discussed below, which are "spaces of maps"; and *cofiber polytopes* that should play the role of "cokernel objects" — for which we lack even a good definition (Problem 9.16*).

Definition 9.16. Let $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$ be full-dimensional polytopes (of dimensions p and q). Then the set of affine maps $f^{A,\boldsymbol{z}} : \mathbb{R}^p \longrightarrow \mathbb{R}^q$, $\boldsymbol{x} \longmapsto A\boldsymbol{x} + \boldsymbol{z}$ can be identified with $\mathbb{R}^{q \times p} \times \mathbb{R}^q \cong \mathbb{R}^{(p+1)q}$. The subset

$$\Phi(P, Q) \quad := \quad \{(A, \boldsymbol{z}) \in \mathbb{R}^{(p+1)q} : f^{A,\boldsymbol{z}}(P) \subseteq Q\}$$

is a polytope in $\mathbb{R}^{(p+1)q}$ of dimension $(p+1)q$: the *mapping polytope* of the pair (P, Q).

We omit the (easy) proof that $\Phi(P, Q)$ actually is a full-dimensional polytope (see [450]). Also there is a natural generalization to polyhedra. Here we note a few important examples.

Examples 9.17.

(i) When P is a point ($p = 0$), then $\Phi(P, Q)$ is isomorphic to Q. More generally, if $P = \Delta_p$ is a simplex with $p+1$ vertices, then the affine image of the vertices of P can be chosen independently in Q, which proves an affine equivalence

$$\Phi(\Delta_p, Q) \quad \cong \quad Q^{p+1}.$$

(ii) In particular, take the $(d-1)$-simplex $\Delta_{d-1} = \mathrm{conv}\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d\} \subseteq \mathbb{R}^d$ on the hyperplane $H := \{\boldsymbol{x} \in \mathbb{R}^d : \mathbb{1}\boldsymbol{x} = 1\}$, as before. Then we can identify affine maps with $H \longrightarrow H$ with linear maps $\mathbb{R}^d \longrightarrow \mathbb{R}^d$ that fix $\mathbf{0}$. Such maps are given by a matrix $V = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d) \in \mathbb{R}^{d \times d}$, where $\boldsymbol{e}_i \longmapsto \boldsymbol{v}_i$. Thus the mapping polytope $\Phi(\Delta_{d-1}, \Delta_{d-1})$ turns out to be

$$\begin{aligned}
\Phi(\Delta_{d-1}, \Delta_{d-1}) \quad &= \quad \{(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d) \in \mathbb{R}^{d \times d} : \ \boldsymbol{v}_i \in \Delta_{d-1} \text{ for } 1 \le i \le d\} \\
&= \quad \{(v_{ij}) \in \mathbb{R}^{d \times d} : \ v_{i,j} \ge 0 \text{ for } 1 \le i, j \le d, \\
&\qquad\qquad \sum_j v_{ij} = 1 \text{ for } 1 \le i \le d\} \\
&= \quad (\Delta_{d-1})^d.
\end{aligned}$$

(iii) Now we will consider the subset of all maps that preserve barycenters, that is, map the barycenter $\frac{1}{d}\mathbf{1}$ of Δ_{d-1} to itself. But the map takes this to $\frac{1}{d}(\boldsymbol{v}_1 + \ldots + \boldsymbol{v}_d)$, so we get extra conditions

$$\sum_{i=1}^{d} v_{ij} = 1, \quad \text{for } 1 \leq i \leq d.$$

Thus the *relative mapping polytope* of all maps from $(\Delta_{d-1}, \frac{1}{d}\mathbf{1})$ to itself turns out to be

$$\Phi((\Delta_{d-1}, \tfrac{1}{d}\mathbf{1}), (\Delta_{d-1}, \tfrac{1}{d}\mathbf{1})) \quad = \quad \Big\{(v_{ij}) \in \mathbb{R}^{d\times d} : \begin{array}{l} v_{ij} \geq 0, \\ \sum_i v_{ij} = 1 \text{ for all } i, \\ \sum_j v_{ij} = 1 \text{ for all } j \end{array} \Big\}$$

— and this is the Birkhoff polytope of all doubly stochastic matrices, as in Definition 0.11.

The full version of a theory of “Universal Constructions for Polytopes” will, I suspect, need two important extensions (both of which correspond to fundamental features in the modern development of algebraic topology):

1. There should be an *equivariant* set-up, which takes into account the study of group actions and symmetries on the polytopes. So, for example, our construction of the permuto-associahedra relies on a subtle interaction of fiber polytopes and a symmetry group action, and this should be collected in a general construction of equivariant fiber polytopes.

2. It should admit *polytope pairs* rather than polytopes as the primary objects: for example, unbounded polyhedra can often be treated in terms of pairs formed by a polytope together with a facet. Also, the maps of simplices $\Delta_p \longrightarrow \Delta_q$ that preserve the barycenter of Example 9.17(iii) fit this pattern, as do those of Exercise 9.15.

Notes

The original motivation for the constructions of secondary polytopes and of fiber polytopes did not come from polytope theory, but from the theory of $\mathcal{A}$-hypergeometric functions [228, 230], and from state polytopes in commutative algebra [58] [535]. It turns out that also there are strong connections to constructions in algebraic geometry [314] and elimination theory [315]. A survey was given by Loeser [367] in the “Bourbaki seminar.” We especially